

SABNOCK WORKS

PROGRESS CHANGES WORK

From the Mechanical Loom to Digital Photography - and What These Upheavals Can Explain About AI



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1. Steam Engine and Loom: New Work, Lost Livelihoods

Steam engines and mechanised textile production changed the world of work. Improved steam engines supplied power to factories, making them less dependent on suitable waterways. In the textile industry, production moved from the homes of domestic workers into large mills. Growing industrial cities needed labour. New employment emerged while established forms of earning a living came under pressure.

The mechanical loom made cloth available faster and at lower prices. For handloom weavers, this was threatening competition: they found it increasingly difficult to compete with the prices of machine-made goods. The Science and Industry Museum documents protests in which looms were destroyed. While factories expanded their production, other people lost the basis of their livelihoods.

The Silesian weavers' revolt of 4 to 6 June 1844 shows what lay behind such conflicts. According to the German Historical Museum, several causes came together: competition from cheaper factory products, an oversupply of labour, falling pay and rising food prices. Exactly as today. The weavers directed their anger at manufacturers and merchants whom they held responsible for their hardship. The Prussian military suppressed the revolt; eleven people were killed.

It would not do justice to their situation to regard these people merely as opponents of progress. Longer working hours could not compensate for the declining value of their work. The advantages of the new production methods offered them little help if they could no longer feed their families from their earnings.

Here lies a parallel with the debate about AI. A technology can improve production and enable new work while individual occupational groups lose income. Acemoglu and Restrepo therefore distinguish between the displacement of existing activities through automation and the creation of new tasks for people. It cannot be taken for granted that the new tasks will offset the losses.

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2. Desktop Publishing: A Printing House Moves onto the Desk

Before newspaper pages were created on screen, compositors had to prepare texts for printing. In traditional hand composition, they assembled individual pieces of type into lines. Later, typesetting machines took over much of this work. The Linotype, developed in 1886, cast entire lines in metal and became an important machine in newspaper production. The daily newspaper still required specialists who turned manuscripts into print-ready type.

This profession changed through several technological upheavals. Hot-metal typesetting was followed by phototypesetting, increasingly supported by computers. Electronic systems were already being used for newspaper production in the 1970s. Desktop publishing continued this development in the 1980s and brought further parts of the production process onto the desk.

One important component was Adobe's PostScript, released in 1984. The language described text and graphics in a way that allowed different output devices to work with them. Together with computers, layout programs and suitable printers, it became possible to combine many things that had previously required separate operations and specialised equipment.

The traditional compositor with a type case and lead type largely disappeared from everyday newspaper production. Some tasks were eliminated, while others moved into digital typesetting, layout and media production. This happened gradually and began long before the breakthrough of desktop publishing. Even so, the result was profound: a familiar craft lost its previous form.

Typographic skill did not become meaningless as a result. A program could output a page; whether it was easy to read and carefully designed remained a question of the decisions made by the person in front of it. At the same time, more people could produce publications themselves without having the equipment of a traditional typesetting shop.

This recalls AI. Tools make tasks accessible that previously required a longer production chain or more specialist knowledge. This changes who can do the work and how much they are paid for it. Easier access, however, does not guarantee good design.

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3. Digital Image Editing: From Technical Experiment to Recognised Tool

Computer art first had to earn its place. The Victoria and Albert Museum describes how little attention the established art world paid to early computer work. One reason was that many of its creators came from science and engineering. They had access to the rare computers of the time but did not fit the familiar image of a trained artist.

To put it pointedly: people saw technology enthusiasts tinkering at their machines rather than artists with a new means of expression. This is a description of the distance felt at the time, not a recorded quotation. There were certainly early exhibitions and supporters. Yet digital work did not automatically receive the same standing as established art forms. The collector Patric D. Prince was among those who recognised its value early and preserved works that might otherwise have been lost.

Programs such as Photoshop, released in 1990, made work on digital images more widely accessible. Artists, photographers and designers could alter and combine images without carrying out every step using analogue methods. Advertising, film and public visual culture were shaped by this development. At the same time, distrust grew toward images whose alteration could not be recognised.

Today, Photoshop is an established tool in professional image design. Digital art is collected, exhibited and studied by museums. The idea that a work cannot be art solely because it was created on a computer

is difficult to reconcile with this practice. The quality of individual works continues to be debated. The fundamental recognition of digital design, however, is now well advanced.

This holds an important reminder for the debate about AI: an unfamiliar tool may initially arouse distrust and later become a natural part of the craft. Artists have already experienced changes in their working tools and in the expectations placed on their work.

The comparison has limits. Photoshop and generative AI intervene in the creation of an image in different ways. Generative systems can create substantial parts of an image and raise additional questions about training rights and authorship. The recognition of digital art does not answer these questions. It does, however, argue against the blanket judgement that new technology must inevitably devalue artistic work.

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4. Digital Photography: More Images, Less Film

Digital photography changed our everyday lives. Images could be viewed, stored and shared immediately. Smartphone cameras intensified this development. No one had to buy film or wait for processing for an additional photograph. Photography increasingly became an everyday form of communication.

For companies that earned money from film, photographic paper and processing, this was a severe disruption. Eastman Kodak and its US subsidiaries filed for protection under Chapter 11 of US bankruptcy law on 19 January 2012. On 3 September 2013, Kodak emerged from the proceedings after restructuring. The company did not disappear, but it had to reposition itself fundamentally.

The familiar explanation that Kodak simply ignored digital photography is too convenient. Kodak documents its own digital developments from 1975 onwards. Vinokurova and Kapoor show that the company invested in new technology and business areas over several decades. Nevertheless, it did not find a permanently viable way to continue its earlier success. Recognising the change and responding to it was not enough.

AgfaPhoto GmbH also became insolvent. Agfa-Gevaert sold its Consumer Imaging business in 2004; the subsequently independent AgfaPhoto GmbH filed for insolvency in 2005. Eurofound documented 1,800 jobs at risk at the time and pointed to the pressure from digital photography. This refers to the divested business, not to the insolvency of the entire Agfa-Gevaert Group.

Fujifilm developed differently. According to the company's own account, it responded to the collapse of the film business by applying existing knowledge to new fields of business. This corporate history does

not independently explain every cause of its success. It does, however, show a deliberate reorientation. The same market disruption produced different results for different companies.

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5. What We Can Learn from This for AI

These upheavals had different causes and consequences. Nevertheless, one conflict keeps returning: new tools make services cheaper or more accessible while people and companies lose their previous positions. At the same time, new tasks emerge. Research into automation examines both developments without assuming that they will automatically balance each other out.

Such shifts are already visible with AI. Studies of the freelance market find declining demand in certain fields of activity and growth in others. This does not yet allow a reliable assessment of the long-term consequences. Nor is it certain that those who lose contracts today will be able to take on the new tasks tomorrow.

Nevertheless, I want to help shape the change. Those who understand the tools can examine their possibilities, recognise their limits and make their own decisions. Anyone who bases their entire strategy on rejecting them gives up part of that ability to act.

We owe those affected more than the statement that they simply have to adapt. Kodak shows that even serious attempts to adapt can fail. The weavers remind us how little a general gain in productivity helps when one's own family has nothing to eat as a result. Progress must also be measured by the future it opens to the people whose work it changes.

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